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A MODEL OF BALANCED CHOICE BEHAVIOR

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In some product categories, consumers appear to alternate among items in a consistent and purposeful, though not entirely deterministic, fashion. Such alternation occurs even in the absence of any apparent external influences on choice. To explain such balanced choice behavior, we present a model based upon the premise that past consumption influences the utility of items on subsequent choice occasions. Individuals may seek variety on certain product attributes and exhibit loyalty toward others. The model specification yields a logit choice model which may be calibrated using consumption diary panel data at the individual or cross-sectional level.

(Variety Seeking; Brand Loyalty; Logit)

1. Introduction

Individuals vary their choice among familiar products for many reasons. These include marketing mix effects (e.g. price change, couponing, advertising pressure), constraints on choice (e.g. stock out, budget limitations, time pressures), and changes in consumption or usage situation (e.g. guests present, multiple product uses). Yet even in the absence of apparent external influences, individuals often switch among alternatives (see Bass, Pessemier and Lehmann 1972, McAlister 1982, McAlister and Pessemier 1982).

Much attention has been devoted to modeling varied choice behavior in marketing. Early work proposed zero-order processes such as Bernoulli (e.g. Frank 1962) and multinomial (e.g. Massy, Montgomery and Morrison 1970, Bass 1974) to describe individual household choice behavior. Recent research (e.g. Givon and Horsky 1978, Blattberg and Sen 1979, Bass et al. 1984) suggests that individuals or households exhibit different types of choice behavior: zero-order and nonzero-order. To accommodate deviations from zero-order behavior, researchers have proposed first-order models of brand choice (Morrison 1966), choice inertia (Jeuland 1979), and choice-event feedback (e.g. Kuehn 1962, Massy, Montgomery and Morrison 1970). These models suggest that an individual is more likely (or increasingly likely, in the case of feedback models) to choose a brand following its recent choice.

In contrast, models of intertemporal variety seeking behavior—deterministic (e.g. McAlister 1982), partially deterministic (Jeuland 1978), and probabilistic (e.g. Jones and Pekelman 1982, Lattin and McAlister 1985)—have also been proposed to explain consumer alternation among familiar items. Underlying these models is the notion (see

McAlister 1982) that individuals experience “satiation” on the attributes provided by available brands, and are therefore *less* likely to choose a brand after its recent choice.

In this research, we propose a model of individual varied choice behavior; i.e., a model to describe the tendency of an individual to make different choices from among familiar items for reasons entirely internal to the individual (c.f. McAlister and Peseemier’s 1982 description of direct, intrapersonal variety seeking behavior). Our goal is to provide a framework capable of capturing the diversity in individual choice behavior identified in previous research (i.e. the effects of positive choice-event feedback and product attribute satiation), in order to gain insight into the process underlying varied choice and to predict such behavior accurately.

The model holds that the variability in individual choice stems partly from the need to *balance* current consumption according to the impact of past consumption; hence, the name “balance model.” As the individual tires of the features or characteristics provided by a recent choice, different alternatives become relatively more appealing. The choice outcome depends upon the individual’s preference for the characteristics provided by each alternative, the lingering impact of past consumption (which we call here the *salience*), and the relationship between salience and the utility of present consumption. The model also recognizes that the choice of an item may have a reinforcing effect on consumer habit, leading to an increased tendency to select the item on a subsequent choice occasion. Thus, the balance model represents individual varied choice behavior as a process which may involve both positive choice-event feedback (not unlike linear learning models) and satiation effects across product characteristics (not unlike McAlister 1982).

We illustrate the balance model by comparison to a first-order model of choice proposed by Givon (1984). Givon’s model describes the average tendency of an individual to seek or avoid variety, either across brands or across partitions (e.g. soft drinks partitioned into cola and noncola categories). In contrast, the balance model simultaneously identifies and differentiates brand-level and attribute-level effects on individual choice. The results suggest a richer set of implications for brand management in understanding consumer response.

The balance model is calibrated using data on individual consumption from diary panel data. Unfortunately, consumption diary panel data are typically available over periods of limited duration, providing relatively few observations per individual. For example, MRCA’s menu census covers a two-week period; even for products consumed on a daily basis, we might expect to find only a dozen observations per person.

To circumvent the often substantial data requirements of individual level models, we offer an approach for calibrating the balance model at a cross-sectional level. To accommodate the heterogeneity in individual taste and reaction to past consumption, we write the balance model parameters as functions of specific individual measures. This approach makes relatively efficient use of data (at some cost to the flexibility of the functional form), and helps to preserve the interpretability of the results at the individual level. Although motivated by somewhat different goals, the result is not unlike the choice model offered by Guadagni and Little (1983); the difference is that the balance model specification permits variety seeking as well as loyalty for brands and attributes.

We conduct our tests of the balance model using data collected by McAlister (1982) on student consumption of soft drinks. These data are relatively free from external influences on choice. Our concern for the internal validity of the calibrated model comes at a sacrifice of external validity: the sample size is small and the results are not generalizable. Therefore, we present our results as a test of the proposed modeling framework on a special segment of consumers, and suggest implications only as illustrative of what might be found in applying the model in a more fully general setting.

2. The Balance Model

We begin by establishing a theoretical framework for individual varied choice, from which we specify our model of balanced choice behavior. By “balance,” we mean the tendency of an individual to weigh the relative utilities offered by each available item in light of the impact of past consumption (note that Farquhar and Rao 1976 use the term somewhat differently, referring to the “attribute balance” exhibited by a collection of items chosen simultaneously). The balancing occurs as the individual shifts his or her pattern of choice, moving away from some characteristics (which may be diminished in desirability due to recent consumption) and toward others in order to derive the highest possible utility from the next consumption occasion.

2.1. Model Development

We begin with the premise that an individual chooses the item that maximizes his or her utility on a given choice occasion. This premise suggests that individuals are myopic, acting in their best self-interest without looking ahead to future opportunities. It also focuses on the problem of conditional choice. Our goal here is not to explain consumption incidence or quantity, but to better understand patterns of choice among familiar alternatives *given* the incidence of choice.

Our second premise holds that the utility of an item is given by the sum of the utilities provided by the constituent characteristics of the item. We use the term “characteristic” to denote both continuous product attributes (e.g. sugar content in soft drinks) and discrete product features (e.g. whether or not a soft drink is a cola). If we further assume that utility may be represented using a compensatory vector model (a special case of the ideal point model described by Green and Srinivasan 1978), then the utility of an item may be written as follows:

$$U_{ic}^h = \sum_p u_{pc}^h x_{pi}, \quad \text{where} \quad (1)$$

U_{ic}^h = the utility of item i for individual h on consumption occasion c ,

x_{pi} = the amount of characteristic p provided by item i (if p represents a discrete product feature, x_{pi} is coded either 0 or 1), and

u_{pc}^h = the utility provided by one unit of characteristic p on consumption occasion c (if p represents a discrete product feature, u_{pc}^h is interpretable as a part-worth utility).

The following empirical tests of the model involve discrete product features (similar to “forms” as used in the context of the Hendry model; see e.g. Kalwani and Morrison 1977). Further testing of the model using continuous attributes remains as a topic for further research.

Our third premise holds that the utility derived from a product characteristic depends upon its prominent or *salient* quality in the memory of the consumer. We use the term “salience” to refer to the lingering or residual impact of a characteristic following its recent consumption. Consider the case in which an individual consumes a cola-flavored, nondiet beverage produced by Pepsi. Immediately following consumption, the characteristics cola, nondiet, and Pepsi are most salient to the consumer (i.e. stand out most prominently). If the individual subsequently consumes a different item (e.g. a noncola, nondiet beverage produced by 7Up), then the salience of cola and Pepsi diminish, overshadowed by the more recent intake of the noncola 7Up. The salience of the nondiet characteristic increases, reinforced by the consumption of a second consecutive nondiet drink.

We represent the salience of characteristic p on choice occasion c as a weighted average of past consumption:

$$s_{pc}^h = \lambda^h s_{pc-1}^h + (1 - \lambda^h) \sum_{j \in A^h} \delta_{jc-1}^h x_{pj}, \quad \text{where} \quad (2)$$

s_{pc}^h = the salience of characteristic p to individual h on consumption occasion c ,
 A^h = the set of acceptable items available to individual h ,
 δ_{jc-1}^h = a variable indicating which item was consumed by individual h on choice occasion $c - 1$ (allowing for the choice of a single item, we require $\sum_j \delta_{jc}^h = 1$), and
 λ^h = a weight in the interval $[0, 1)$ reflecting the relative impact of salience at $c - 1$ on salience at c .

The exponential smoothing model in (2) enables us to represent the salience of characteristic p as the average amount of p consumed on previous occasions, with the most recent consumption events weighted most heavily. The use of an exponential form to describe the dynamic impact of past consumption is parsimonious and has relatively strong precedent (see Jeuland 1978, McAlister 1982).

According to our third premise, u_{pc}^h is a function of s_{pc}^h , which may have either a positive or negative effect. McAlister (1982) finds empirical support for attribute satiation in soft drinks, suggesting that as the salience of an attribute increases, the marginal utility of an additional unit of that attribute diminishes. Givon (1984) finds evidence of widespread variety avoidance behavior in categories such as facial tissues, aluminum foil and liquid detergent, suggesting that the overall effect of past consumption is to increase the utility of the item.

Writing u_{pc}^h as a simple linear function of s_{pc}^h and substituting into equation (1) yields:

$$U_{ic}^h = \sum_p (w_p^h + v_p^h s_{pc}^h) x_{pi}, \quad \text{where} \quad (3)$$

w_p^h = an intercept term, interpretable as the utility of characteristic p at zero salience, and

v_p^h = a slope term, interpretable as the impact of a unit change in salience on the utility of characteristic p .

If $v_p^h < 0$, we say that individual h seeks variety on characteristic p . If $v_p^h > 0$, we say that individual h exhibits loyalty to characteristic p .

Equation (3) describes the utility of item i in terms of past consumption activity and its impact upon characteristics which are common to all items. An individual may also derive utility from tangible or intangible characteristics unique to a specific item. For example, an individual may alternate between colas and noncolas, and yet choose Coke over Pepsi consistently. The apparent loyalty to Coke (when choosing a cola) may be due to a preference for the unique image associated with the brand itself. We capture this item-specific effect by modifying equation (3) as follows:

$$U_{ic}^h = (W_i^h + V^h S_{ic}^h) + \sum_p (w_p^h + v_p^h s_{pc}^h) x_{pi}, \quad \text{where} \quad (4)$$

W_i^h = an intercept term, interpretable as the part-worth utility of item i at zero salience. In terms used by Srinivasan (1979), W_i^h is an item-specific constant that captures the unique aspects of item i such as brand image, etc.

V^h = a slope parameter, describing the impact of a unit change in item salience on the part-worth utility of item i .

$$S_{ic}^h = \lambda^h S_{ic-1}^h + (1 - \lambda^h) \delta_{ic-1}^h$$

= the salience of item i to individual h on occasion c . Note that our proposed form for item salience is identical to the exponential smoothing model used by Guadagni and Little (1983) as their construct of brand loyalty.

Returning to our first premise, an individual chooses the item that maximizes his or her utility on a given choice occasion. If we assume that the nonsystematic factors influencing utility (denoted ϵ_{ic}^h) are distributed independently and identically according to the double-exponential distribution, then the probability of choice for item i is given by the multinomial logit form:

$$p_{ic}^h = \exp(U_{ic}^h) / \sum_{k \in A^h} \exp(U_{kc}^h). \quad (5)$$

There are two possible concerns with imposing the logit assumptions. First, it may not generally be true that ϵ_{ic}^h is independent of ϵ_{ic}^h . This problem, a violation of the independence of irrelevant alternatives (IIA), raises the concern that the model will yield systematic errors in the predicted choice probabilities. Second, it may not generally be true that ϵ_{ic}^h is independent of ϵ_{ic-1}^h . In addition to the IIA problem, this raises the additional concern about the efficiency of the estimation. Despite these limitations, the logit model provides a reasonable, computationally tractable form for relating utility to probability of choice.

The parameters of the logit model can be estimated using a gradient search procedure to maximize the likelihood function. By specifying λ^h *a priori*, the utility function becomes linear in the parameters, guaranteeing convexity of the likelihood function. Although it would be possible to search over all values of $\lambda^h \in [0, 1)$ for a globally optimal solution, this may lead to overfitting when dealing with individual consumption histories that are short in length.

With the logit form, it is possible to estimate at most $(n - 1)$ of the intercept terms W_i^h and w_p^h , where n is the number of different items in the choice set. The use of n or more constant terms leads to a singularity in the estimation (see also Guadagni and Little 1983). Thus, the number of parameters required by the balance model is $(n - 1)$ plus one for item salience (V^h) plus one each for the salience of characteristic p (v_p^h).

2.2. Contrast to Existing Model

It is instructive to compare the final form for the balance model to an existing model of variety seeking behavior proposed by Lattin and McAlister (1985). Assuming $\lambda^h = 0$ (i.e. salience on occasion c is completely determined by consumption occasion $c - 1$) and substituting equation (4) into (5), we can write from the balance model the probability that individual h will choose item i following the consumption of item j as

$$p_{ic}^h = \frac{\exp\{(W_i^h + \sum_p w_p^h x_{pi}) + (\sum_p v_p^h x_{pj} x_{pi})\}}{\sum_k \exp\{(W_k^h + \sum_p w_p^h x_{pk}) + (\sum_p v_p^h x_{pj} x_{pk})\}} \quad (6)$$

and compare it to the first-order transition probability proposed by Lattin and McAlister:

$$P_{ij}^h = \frac{\pi_i^h - V^h S_{ij}^h}{\sum_k (\pi_k^h - V^h S_{kj}^h)} \quad \text{where} \quad (7)$$

π_i^h = the value to individual h of all features provided by item i ,

S_{ij}^h = the value to individual h of all features shared by items i and j , and

V^h = a discount factor representing the variety-seeking intensity of individual h .

There are two noteworthy differences between the models in (6) and (7). First, Lattin and McAlister assume that variety seeking is an individual level phenomenon, described by a single parameter V^h . Thus, if an individual desires variety, he or she seeks it uniformly across all characteristics provided by the available items. The balance model, on the other hand, assumes that an individual's desire for variety may differ across characteristics. Second, Lattin and McAlister's model requires the estimation of S_{ij}^h for all possible pairs of items in the choice set. S_{ij}^h reveals how much individual h values the features shared by items i and j , but it does not say which features the items share or how much each feature is valued. In contrast, the balance model captures the impact of product characteristics specified *a priori*.

3. Illustration

We illustrate the balance model by comparison to another variety seeking model, proposed by Givon (1984). Unlike McAlister (1982) and Lattin and McAlister (1985), Givon modeled variety seeking behavior on a continuum ranging from extreme desire for change to extreme desire to avoid change. Using the parameter $VS \in [-1, 1]$ to represent the variety seeking/avoidance behavior of the individual or household, Givon proposed the following model:

$$P_{ij} = (|VS| + VS)/(2n - 2) + (1 - |VS|)\theta_j, \quad i \neq j, \quad \text{where} \quad (8)$$

P_{ij} = the probability of choosing item j following a choice of item i (by definition $P_{jj} = 1 - \sum_{k \neq j} P_{kj}$),

n = the number of items in the individual's choice set, and

θ_j = a parameter in the interval $[0, 1]$ representing the preference of the individual for item j in the absence of variety seeking/avoidance.

When $VS = 1$, the desire for variety becomes so great that the individual will choose any item other than i with equal probability, regardless of preference θ_j . Givon recognized this as a restrictive assumption, and suggested that some individuals might be better described as seeking/avoidance variety across product partitions. Givon recommended estimating a parameter VP (analogous to VS) for each of several different partitions, and comparing the results to find the best representation of behavior.

In contrast to Givon's approach, the balance model can be used to capture simultaneously the individual's desire to seek/avoid variety across several product characteristics. Thus, v_p^h represents the *net* tendency for individual h to seek/avoid variety on characteristic p , *controlling for the effects of other characteristics*. The balance model therefore has the potential for determining which of two correlated attributes provides the principal motivation for varied choice. For example, an individual seeking variety in texture among unsweetened cereals (e.g. flakes vs O's) might be seen as seeking variety in grain (e.g. corn vs. wheat). To the extent that the attributes are not perfectly correlated, the balance model is in principle capable of separating the two effects.

3.1. Data

In order to provide a valid test of the balance model, we must have data at the individual level. Unfortunately, this rules out many existing sources of panel data collected at the household level. As shown by Kahn, Morrison and Wright (1985), data collection at the household level "obscures the observation of some potentially interesting individual-level behavior." In fact, with typical panel data it is impossible to know whether choice variation is due to individual balanced choice or to alternating choices by two zero-order individuals.

We must also rule out purchase panel data (e.g. IRI scanner panels), not only because the data are at the household level, but also because the motivation for choice is not oriented toward immediate consumption. At the point of purchase, the shopper may be induced to purchase because of favorable price or special display. When the family pantry is partially or fully stocked, the pattern of purchase behavior need not follow directly patterns of consumption within the household.

One possible source of appropriate data is a menu census, such as those conducted by MRCA. For each of 2000 households, MRCA collects detailed diary information on all food consumed by members of the household over a two-week period. Each diary includes information on every food item served in the home or consumed away from home, the date and time of each consumption occasion, and which members of the household consumed the item(s). From these diaries, it would be possible to create individual consumption histories for specific product categories (e.g. soft drinks). Un-

fortunately, without more than two weeks of data, the analysis would have to be limited to only the most frequently consumed products.

Another possible source of data is a consumption diary following a new product pre-test procedure such as ASSESSOR (Silk and Urban 1978). Following some ASSESSOR studies, individuals are recruited to keep a usage/consumption diary for a specific class of products, recording the time and date of each consumption event. Because keeping track of a single product category is less arduous than the MRCA menu census, it might be possible to ask the ASSESSOR panelist to provide additional information over a longer period of time. Knowing which items are unavailable in the household (and perhaps which are available as leftovers) on each consumption occasion would be helpful in controlling for external effects on varied choice behavior.

A third possible source of data is the independent research study. To illustrate the use of the balance model, we use data on student consumption of soft drinks collected by McAlister (1982). McAlister's data enables us to examine the variability in individual choice due to internal factors, which is the goal of the balance model. We do not claim generalizability for these data. Nonetheless, they afford an opportunity to illustrate the balance model and the type of insights it provides.

In McAlister's study, 36 students recorded on a semiweekly basis each consumption of a soft drink over an entire term (79 days). For this study, we dropped any drink with less than a 3% choice share as too small to permit reliable estimation. The seven remaining soft drink brands most frequently consumed by the students were: (1) Coke, (2) Pepsi, (3) 7Up, (4) Tab, (5) Diet Pepsi, (6) Dr. Pepper, and (7) Sprite. Of the 36 students, 23 made choices almost exclusively from this set of drinks, reporting 667 out of a total of 698 collective choices from the seven (an average of 29 drinks per student). For the purposes of our illustration and comparison with Givon's model, we limited our analyses to the nine individuals who made at least 20 choices from among colas (Coke, Pepsi, Tab, Diet Pepsi, Dr. Pepper) and noncolas (7Up, Sprite).

3.2. Model Calibration

We calibrate two specifications of Givon's model: one based on variety seeking/avoidance among items (in which we estimate VS) and one based on variety seeking/avoidance between colas and noncolas (in which we estimate VP). Given the first choice observation to initialize the switching process, we maximize the following conditional likelihood function:

$$LL(\text{Givon}) = \sum_{i=1}^n \sum_{j=1}^n n_{ij} \ln(P_{ij}), \quad (9)$$

where n is the number of items in the individual's choice set and n_{ij} is the number of choice transitions from item i to item j . We then test each specification of Givon's model against the zero-order model using the likelihood ratio test (distributed χ^2 with one degree of freedom).

For the purposes of illustration, we also calibrate two specifications of the balance model: one in which we specify $\lambda^h = 0.0$ and one in which $\lambda^h = 0.5$. In each specification, we estimate simultaneously the parameters V^h (describing the impact of the salience of an item on its utility) and v_1^h (describing the impact of the salience of the cola/noncola characteristic on its utility). Given the first choice observation to initialize salience, we maximize the following conditional likelihood function:

$$LL(\text{Balance}) = \sum_{\substack{c \in C^h \\ c \neq 1}} \sum_{i \in A^h} \delta_{ic}^h \ln(p_{ic}^h), \quad (10)$$

where C^h is the set of consumption occasions for individual h . We then test the result

from each specification against the zero-order model using the likelihood ratio test (distributed χ^2 with two degrees of freedom).

The results are provided in Table 1. Based on the likelihood ratio tests, about half of the nine subjects (five according to Givon, four according to the balance model) exhibit behavior that can be statistically differentiated from zero-order behavior at the 0.10 level. Note that for every subject, the balance model (with either $\lambda^h = 0.0$ or $\lambda^h = 0.5$) provides at least as good a fit as either of the specifications of Givon's model (across items or between cola/noncola partitions). This is reassuring, but hardly surprising since the balance model includes an additional parameter.

An interesting difference between the two models is the pattern of parameters describing individual varied choice behavior. In Givon's models, the sign of VS is the same as the sign of VP for eight of the nine subjects. Thus, we might conclude that an individual who seeks variety across soft drink items will also be likely to seek variety between colas and noncolas. In contrast, the results from the balance model with $\lambda^h = 0.0$ suggest that seven out of nine individuals exhibit *different* signs for the parameters V^h and v_1^h . For example, subject 7 exhibits loyalty in his selection of different soft drink items ($V^h = 0.88$) but apparently desires some variety between colas and noncolas ($v_1^h = -0.27$). By estimating V^h and v_1^h simultaneously, we are able to identify a tendency to switch between colas and noncolas while controlling for the tendency to remain loyal to the last item chosen. In Givon's model, subject 7's apparent item loyalty ($VS = -0.20$) may be influencing the estimated parameter value of VP , which describes him as someone exhibiting loyalty toward cola or noncola ($VP = -0.18$).

Because of the nature of the data, our comparison of the balance model to Givon's model is intended only to illustrate the kinds of differences we might expect to find in a more fully general situation. We are still faced with several questions. In particular, can we characterize the behavior of less frequent consumers (i.e. the 14 subjects with fewer than 20 consumption occasions)? And, can we show that the balance model not only fits but also predicts choice better than an established model?

4. Cross-Sectional Model

What makes it difficult to answer these questions is the lack of sufficiently long individual choice histories. Most consumption histories do not provide enough observations to permit reliable calibration and predictive validation of the balance model. In an attempt to circumvent the data constraints encountered at the individual level, we offer the following cross-sectional specification of the model. We begin by dividing the panel data (for all individuals) into three periods: initialization, calibration, and validation. We use the data from the first period to take measures of past choice behavior for each individual. We then write the parameters of the balance model (W_i^h, w_p^h, V^h, v_p^h) as functions of these measures, which enables us to preserve some of the heterogeneity in preferences and varied choice behavior across individuals. Using the data from the second period, we then calibrate the cross-sectional balance model. Finally, using the results from the calibration, we attempt to predict choice behavior in the third period.

4.1. Specification

Before we rewrite the model parameters, we need to reconsider the appropriate form for salience in a cross-sectional model. At the individual level, the only source of variance in s_{pc}^h is the past consumption activity of individual h . At the cross-sectional level, however, salience also reflects any variance in consumption activity *across* individuals. To preserve the interpretation of the parameters V^h and v_p^h as the impact of a

TABLE 1
Comparison of Givon Model and Balance Model

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Subject Number	# of Choices	# of Items Chosen	LL for Zero-Order Model	Givon Model				Balance Model					
				Across Items		Across Partitions		V^h	$\lambda^h = 0.0$		V^h	$\lambda^h = 0.5$	
				V/S	LL	VP	LL		v_t^h	LL		v_t^h	LL
1	25	3	-15.28	-0.35	-13.69*	-0.44	-13.01**	-20.98	21.93	-12.80*	0.19	0.86	-14.07
2	33	3	-14.83	0.14	-14.47	0.07	-14.63	-16.48	4.16	-14.10	-7.69	8.27	-13.57
3	22	5	-30.41	0.46	-29.99	0.10	-30.29	-0.99	0.33	-29.48	-2.21	1.82	-29.11
4	40	4	-17.06	0.00	-17.05	0.08	-16.81	-19.29	-2.24	-16.60	-2.88	0.77	-16.77
5	28	5	-33.39	-0.17	-32.25	-0.14	-33.13	1.01	-0.72	-31.92	1.12	-0.93	-32.69
6	46	4	-51.73	0.30	-50.10*	0.42	-47.45***	-1.24	-0.27	-46.00***	0.40	-1.63	-49.42*
7	63	5	-73.55	-0.20	-68.64***	-0.18	-72.59	0.88	-0.27	-70.78*	2.35	-1.39	-67.91***
8	27	3	-23.69	-0.16	-23.34	-0.38	-21.76**	-21.56	22.11	-19.77**	-0.79	2.01	-20.51*
9	53	4	-42.37	-0.19	-40.41**	0.06	-42.19	0.79	-9.93	-40.63	1.35	-0.50	-40.37

Recall, V/S, VP > 0 indicate desire for variety.

* Significantly different from zero-order at 10%; $\chi^2(0.10) = 2.71$

** Significantly different from zero-order at 5%; $\chi^2(0.05) = 3.84$.

*** Significantly different from zero-order at 1%; $\chi^2(0.01) = 6.64$.

Recall V^h , $v_t^h < 0$ indicates a desire for variety.

* Significantly different from zero-order at 10%; $\chi^2(0.10) = 4.61$.

** Significantly different from zero-order at 5%; $\chi^2(0.05) = 7.82$.

*** Significantly different from zero-order at 1%; $\chi^2(0.01) = 9.21$.

change in past consumption on individual utility, we must eliminate the source of inter-individual variance in salience.

One way to do this is to rewrite salience as a *relative* construct, expressed as a deviation from the mean of each individual. Since $E(S_{ic}^h)$ is just the expected proportion of the time individual h will choose item i over all choice occasions c , and $E(s_{pc}^h)$ is the expected amount of characteristic p consumed by individual h on any given choice occasion, we can write:

$$RS_{ic}^h = S_{ic}^h - m_i^h \quad \text{and} \quad (11)$$

$$rs_{pc}^h = s_{pc}^h - \sum_j m_j^h x_{pj}, \quad \text{where} \quad (12)$$

RS_{ic}^h = the relative salience of item i to individual h at choice occasion c ,

rs_{pc}^h = the relative salience of characteristic p to individual h at choice occasion c , and

m_i^h = an estimate of the true proportion of times individual h chooses item i , given by the actual proportion of times individual h chose item i during the initialization period.

We now turn to the problem of specifying W_i^h and w_p^h . Using relative salience in the model, W_i^h represents individual h 's part-worth utility for item i at zero relative salience (i.e. average salience). All other things being equal, we would expect a higher part-worth utility to lead individual h to choose item i a greater proportion of the time. Thus, we might write W_i^h as a simple linear function of the measure m_i^h (i.e. $W_i^h = W'_i + W''m_i^h$) to account for some of the variance in idiosyncratic preference. However, due to differences in individual consumption frequency, the measures m_i^h may be based on very different numbers of observations during the initialization period. A measure based on only one consumption occasion will exhibit considerably higher error variance than a measure based upon 10 consumption occasions. The coefficient of m_i^h should differ across individuals, to reflect the stability of the measure. We therefore propose the following expressions for W_i^h and w_p^h :

$$W_i^h = W'_i + (W'' + W'''n^h)m_i^h, \quad (13)$$

$$w_p^h = w'_p + (w''_p + w'''_pn^h) \sum_j m_j^h x_{pj}, \quad (14)$$

where n^h is the number of consumption occasions for individual h during the initialization period. As n^h increases, m_i^h becomes a better, more reliable measure of individual preference, so we expect W''' , $w'''_p > 0$. If we are successful in capturing the heterogeneity in individual preference, we would also expect $(W'' + W'''n^h) > 0$ and $(w''_p + w'''_pn^h) > 0$.

Finally, we address the problem of specifying V^h and v_p^h . The parameter v_p^h represents the impact of a change in salience on the utility of characteristic p . All other things being equal, we would expect an individual with a high desire for variety on characteristic p (i.e. $v_p^h < 0$) to exhibit relatively high variance in his or her consumption of the characteristic (e.g. colas half of the time and noncolas half of the time). On the other hand, an individual loyal to characteristic p (i.e. $v_p^h > 0$) should exhibit much less variance in past consumption. To capture the heterogeneity in desire for variety across individuals, we propose the following expressions for v_p^h and V^h :

$$V^h = V' + V''\Sigma_i^h, \quad (15)$$

$$v_p^h = v'_p + v''_p\sigma_p^h, \quad \text{where} \quad (16)$$

$$\Sigma_i^h = (m_i^h(1 - m_i^h))^{1/2}$$

= an estimate of the true variability of individual consumption of item i , given by the standard deviation in consumption of item i during the initialization period, and

$$\sigma_p^h = (\sum_j m_j^h x_{pj}^2 - (\sum_j m_j^h x_{pj})^2)^{1/2}$$

= an estimate of the true variability of individual consumption of characteristic p ,

given by the standard deviation in consumption of characteristic p during the initialization period.

If we are successful in capturing the heterogeneity in idiosyncratic desire for variety, we would expect V'' , $v_p'' < 0$.

Replacing salience with relative salience and substituting equations (13)–(16) into equation (4) yields the complete specification of the cross-sectional balance model (CSB):

$$U_{ic}^h = W'_i + (W'' + W'''n^h)m_i^h + (V' + V''\Sigma_i^h)RS_{ic}^h \\ + \sum_p \{w'_p + (w''_p + w'''_pn^h)(\sum_j m_j^h x_{pj}) + (v'_p + v''_p\sigma_p^h)rs_{pc}^h\}x_{pi}. \quad (17)$$

We calibrate the model by choosing parameters W'_i , W'' , W''' , V' , V'' , w'_p , w''_p , w'''_p , v'_p , and v''_p to maximize the following likelihood function across all individuals:

$$LL(\text{CSB}) = \sum_{h \in H} \sum_{c \in C^h} \sum_{i \in A} \delta_{ic}^h \ln \{ \exp(U_{ic}^h) / \sum_{k \in A} \exp(U_{kc}^h) \} \quad \text{where} \quad (18)$$

H = the set of individuals in the panel, and

A = the set of items available for consumption (assumed to be the same for all individuals).

For CSB with N items and P characteristics, the number of parameters required is $(N - 1) + 4(P + 1)$.

4.2. Limitations

The cross-sectional specification does not come without some cost to accuracy and functional flexibility. Because we have no strong priors to relate λ^h to observed consumption behavior, we calibrate the cross-sectional model on the assumption that $\lambda^h = \lambda$ for all individuals h (similar approaches are used by McAlister 1982 and Guadagni and Little 1983). Also, the model must be based upon objectively determined characteristics (x_{pi}) as opposed to subjective individual assessments.

A more serious limitation concerns the use of the measures of individual consumption behavior taken in the initialization period. Because we are dealing with choice histories having relatively few observations (necessitating the cross-sectional approach), the measures m_i^h may be based on very few data points. This is potentially a problem for two reasons. First, too much sampling error variance in m_i^h will make it a poor measure for capturing heterogeneity in preference across individuals. Second, unstable values of m_i^h make poor measures of $E(S_{ic}^h)$, which undermines the construct of relative salience. If RS_{ic}^h is not properly expressed relative to average behavior for the individual, then it will continue to reflect to some extent the heterogeneity in preference across individuals. This may exert some positive bias on the coefficients V' and v_p' .

Because of the importance of good measures m_i^h , we suggest that the available panel data be divided equally between initialization and calibration periods. Ideally, for each individual there should be sufficient data so that the initialization period contains more choice observations than there are items in the individual's choice set. Also, at a minimum, the panel should be large enough to provide several hundred choice observations in the calibration period with which to estimate the parameters for the cross-sectional model.

4.3. Model Testing

We illustrate the calibration and testing of the cross-sectional balance model using the McAlister data described in §3 above. To provide for the initialization, calibration and testing of the model, we split the data for the 23 student panelists into three periods.

The first four weeks (28 days), during which time the students make a total of 241 choices of soft drinks, form the initialization period for the model. These data are used only to calculate the individual-specific measures (m_i^h , n^h) and to initialize relative salience (RS_{ic}^h and rs_{pc}^h). The second four weeks, in which 247 choices are made, form the calibration period, while data from the remaining period (23 days, 179 choices) are set aside for a predictive validation of the calibrated model.

The seven soft drinks in the study are often thought of according to two prominent characteristics. Each brand is either a nondiet or diet beverage, and each is either a cola or a noncola. Thus, we introduce the following definitions for characteristics $p = 1$ and $p = 2$:

x_{1i} = a variable indicating whether item i is ($x_{1i} = 1$) or is not ($x_{1i} = 0$) a nondiet beverage, and

x_{2i} = a variable indicating whether item i is ($x_{2i} = 1$) or is not ($x_{2i} = 0$) a cola.

Before we can calibrate the model, we must also specify a value for the exponential smoothing parameter λ used to calculate relative salience. Putting finetuning issues aside, we choose the value of $\lambda = 0.7$ *a priori*. Subsequent model runs with values of 0.6 and 0.8 suggested that the model is not sensitive to small changes in this parameter, and the results reported below are based on the original value of $\lambda = 0.7$. With seven items and two characteristics, the number of parameters required by the model is $(7 - 1) + 4(2 + 1) = 18$.

We assess the fit of the cross-sectional balance model (CSB) by comparing it to a cross-sectional model of brand loyalty. The brand loyalty model (BL) is so named because of its similarity to the choice model proposed by Guadagni and Little (1983). In the absence of direct marketing activity at the point of purchase, Guadagni and Little's model for utility may be written:

$$U_{ic}^h = W_i' + V'(RS_{ic}^h + m_i^h). \quad (19)$$

In terms used by Guadagni and Little, W_i' is a brand-specific constant and $(RS_{ic}^h + m_i^h)$ measures the brand loyalty of individual h to item i on choice occasion c .

We propose a slightly less restrictive version of Guadagni and Little's formulation, in which we allow the coefficient for m_i^h to be different from V' , as follows:

$$U_{ic}^h = W_i' + W''m_i^h + V'RS_{ic}^h. \quad (20)$$

This enables us to express BL as a nested choice model, and to assess the improvement in fit of CSB over BL using a likelihood ratio test (distributed χ^2 with 10 degrees of freedom).

Table 2 shows the fit statistics and parameter estimates for BL and CSB. The U^2 measure (Hauser 1978) indicates that the balance model succeeds in fitting 7.5% of the uncertainty unexplained by the brand loyalty model, and the likelihood ratio test shows that CSB provides an improvement in fit over BL which is significant at the 0.001 level.

Before we discuss the parameter estimates in Table 2, we present the results of a predictive test of CSB against BL. The test involves using the calibrated models to predict probabilities of choice during the validation period. Like Guadagni and Little (1983), we adopt a Monte Carlo procedure for prediction, in which we use the predicted probabilities on consumption occasion c to simulate a choice and then use the simulated choice to update relative salience for consumption occasion $c + 1$. In this way, none of the data from the calibration period enter into the forecast for the validation period. We run the Monte Carlo five times, each time simulating a distinct series of choices. We then average these five series of predicted probabilities (*not* the series of 0/1 choices generated by the simulation) and use this averaged series as the forecast proba-

TABLE 2
Parameter Estimates and Fit Statistics for Brand Loyalty and Cross-Sectional Balance Models

	Brand ^a Loyalty	Balance ^b Model	Balance ^b Model
Item-Specific Characteristics			
W'_1 (Coke)	1.25 (3.47) ^c	1.49 (3.98)	1.28 (4.21)
W'_2 (Pepsi)	0.76 (1.87)	0.78 (1.89)	0.95 (2.86)
W'_3 (7Up)	0.76 (1.76)	0.97 (2.24)	0.91 (2.96)
W'_4 (Tab)	1.01 (2.51)	1.30 (2.83)	0.83 (2.43)
W'_5 (Diet Pepsi)	0 ^d	0 ^d	0 ^d
W'_6 (Dr. Pepper)	0 ^d	0 ^d	0 ^d
W'_7 (Sprite)	0 ^d	0 ^d	0 ^d
W''	5.02 (14.56)	2.24 (3.50)	2.53 (5.00)
W'''		0.10 (2.53)	0.09 (2.90)
V'	2.21 (5.19)	1.24 (0.94)	2.15 (3.22)
V''		0.44 (0.33)	0.22 (0.12)
Nondiet/Diet Characteristic			
w'_1	0.01 (0.03)	-1.71 (-2.37)	-1.79 (-3.14)
w''_1		1.76 (1.49)	1.82 (1.84)
w'''_1		0.10 (1.26)	0.11 (1.43)
v'_1		3.77 (0.94)	4.98 (2.79)
v''_1		-14.16 (-1.10)	-10.93 (-1.77)
Cola/Noncola Characteristic			
w'_2	-0.32 (-0.62)	0.81 (0.85)	-0.16 (-0.23)
w''_2		-0.85 (-0.76)	-0.19 (-0.23)
w'''_2		-0.02 (-0.46)	-0.00 (-0.13)
v'_2		2.51 (1.15)	1.37 (1.06)
v''_2		-7.57 (-1.38)	-5.25 (-1.55)
Fit Statistics			
Log likelihood	-246.2	-227.8	
U^2		0.075	
Likelihood			
Ratio Test		$\chi^2_{10} = 36.8$	
Significance of χ^2		$p < 0.001$	

^a Based on 247 choice observations from calibration period.
^b Based on 426 choice observations from combined calibration and validation periods.
^c *t*-statistic in parentheses.
^d Constrained to zero to avoid singularity in estimation.

bilities. By averaging, we are able to avoid potential distortion from a single “unlikely choice” generated by the Monte Carlo procedure.

We assess the predictions given by BL and CSB in two different ways. First, we calculate from the predicted choice probabilities the log likelihood of the actual choice sequence in the prediction period. This corresponds to the same measure used to assess the fit of each model in the calibration period. Second, we calculate from the predicted choice probabilities the aggregate choice share for each of the seven soft drinks. We then compare the actual aggregate shares to the predicted shares, using the formula for the standard error given below:

$$SE_i = (1/M)(\sum_{h \in H} \sum_{c \in C^h} p_{ic}^h (1 - p_{ic}^h))^{1/2}$$

(21)

where SE_i is the standard error for item i and M is the total number of choices made by all individuals in the validation period. Table 3 presents these comparisons of the models’ predictive performances.

On the basis of log likelihood, the performance of CSB is substantially better than BL when predicting using Monte Carlo. We attribute this difference to the fact that the calibrated brand loyalty model (in which $V' = 2.21$, $t = 5.19$) is strictly a positive choice-event feedback model, in the sense that past experience with an item always increases the probability of choosing the item again. In simulating the model over a number of choice occasions, relative salience moves toward a positive or negative extreme and the predictions for each individual ultimately gravitate toward the repeated choice of a single item. The balance model, on the other hand, recognizes that for some individuals, increasing experience with certain product characteristics decreases the probability of subsequent choice. Choice predictions for these individuals

TABLE 3
Predictive Performance of the Cross-Sectional Balance Model Versus Brand Loyalty Model

Log Likelihoods for Prediction Using Monte Carlo			
	Brand Loyalty		Balance Model
Log Likelihood	−209.6		−179.4
Aggregate Choice Shares for Prediction Using Monte Carlo			
	Brand Loyalty	Balance Model	Actual Share
Coke	0.446*	0.391	0.346
(2 std. error)	(0.053)	(0.054)	
Diet Pepsi	0.122	0.121	0.140
(2 std. error)	(0.035)	(0.024)	
Dr. Pepper	0.034	0.028	0.050
(2 std. error)	(0.025)	(0.024)	
Pepsi	0.110	0.118	0.134
(2 std. error)	(0.037)	(0.037)	
7Up	0.136*	0.189	0.207
(2 std. error)	(0.046)	(0.051)	
Sprite	0.031	0.038	0.039
(2 std. error)	(0.026)	(0.028)	
Tab	0.121	0.115	0.084
(2 std. error)	(0.040)	(0.032)	

* Predicted share is more than 2SE away from actual.

should not gravitate toward a single brand when the model is simulated over several choice occasions. To the extent that nonloyalty better describes the behavior of these individuals, the balance model provides a more accurate prediction.

Table 3 also compares the predictions by BL and CSB on the basis of predicted aggregate choice share. Both models have difficulty in capturing the relatively large drop in the choice share of Coke and the increase in the share of 7Up, although the balance model fares somewhat better than brand loyalty. While the apparent difference between the two is not significant, the accuracy of CSB appears to be quite acceptable.

4.4. Discussion of Parameter Estimates

Having tested the cross-sectional balance model with relatively favorable results, we combined the data from the calibration and validation periods and recalibrated the balance model over 426 choice occasions. These parameter estimates (presented in the third column of Table 2) are used in the following discussion. Recall that these results pertain to a sample of 23 students. Therefore, the discussion below is presented as illustrative of the kind of conclusions which might be drawn from results based on a statistically representative sample.

The coefficient for m_i^h is positive and significant ($W'' = 2.53, t = 5.00$), demonstrating the importance of accounting for heterogeneity in preference among items. Because most individuals drink either diet or nondiet beverages (but not both), an individual's revealed preference for nondiet beverage (given by $\sum_j m_i^h x_{1j}$) helps to account for heterogeneity in preference for the characteristic ($w_1' = 1.82; t = 1.84$). On the other hand, most individuals drink both colas and noncolas, and the effect of revealed preference for cola is not significant ($w_2' = -0.19; t = -0.23$).

The number of observations from the initialization period used to measure m_i^h (given by n^h) plays a significant role in scaling the impact of revealed preference on utility. All other things remaining equal, there is a 0.09 increase in the value of $W_i^h (= W_i' + (W'' + W'''n^h)m_i^h)$ for each unit increase in n^h . The inclusion of n^h as an interaction term enables the model to vary the value of W_i^h according to the precision of the measure m_i^h . For example, if an individual's revealed preference for Coke is $m_1^h = 1.0$ based on one consumption occasion in the initialization period, then $W_1^h = 1.28 + \{2.53 + 0.09(1.0)\}(1.0) = 3.90$. If, however, the same measure is based on 10 consumption occasions, then $W_1^h = 4.71$.

The brand loyalty model suggests that the recent consumption of an item has a positive impact on the probability of its subsequent choice ($V' = 2.21, t = 5.19$). The same result is suggested by the balance model ($V' = 2.15, t = 3.22; V''$ not significant). Thus, individuals appear to experience "choice inertia" across items. Viewed in isolation, this result seems counterintuitive, since previous research (McAlister 1982 and Givon 1984) leads us to expect to see variety seeking among soft drinks. However, it is necessary to extend our view to include the impact of the characteristics provided by the soft drinks. From Table 2, we can see that both v_1^h and v_2^h vary inversely with individual consumption variability ($v_1' = -10.93, t = -1.77; v_2' = -5.25, t = -1.55$). These results suggest that there is fairly substantial heterogeneity in individual variety seeking/avoidance on these two dimensions.

Figure 1 shows the distributions of v_1^h and v_2^h . Among these 23 students, there appears to be very little desire for variety between nondiet and diet beverages: all but one subject exhibits fairly strong loyalty (i.e. $v_1^h > 0$). Nearly half of the students, on the other hand, appear to be seeking variety between colas and noncolas (i.e. $v_2^h < 0$). This result is quite consistent with the findings presented by Givon (1984, Table 7), in which he finds 47% of the subjects in a laboratory experiment exhibited variety seeking behavior between cola and noncola partitions.

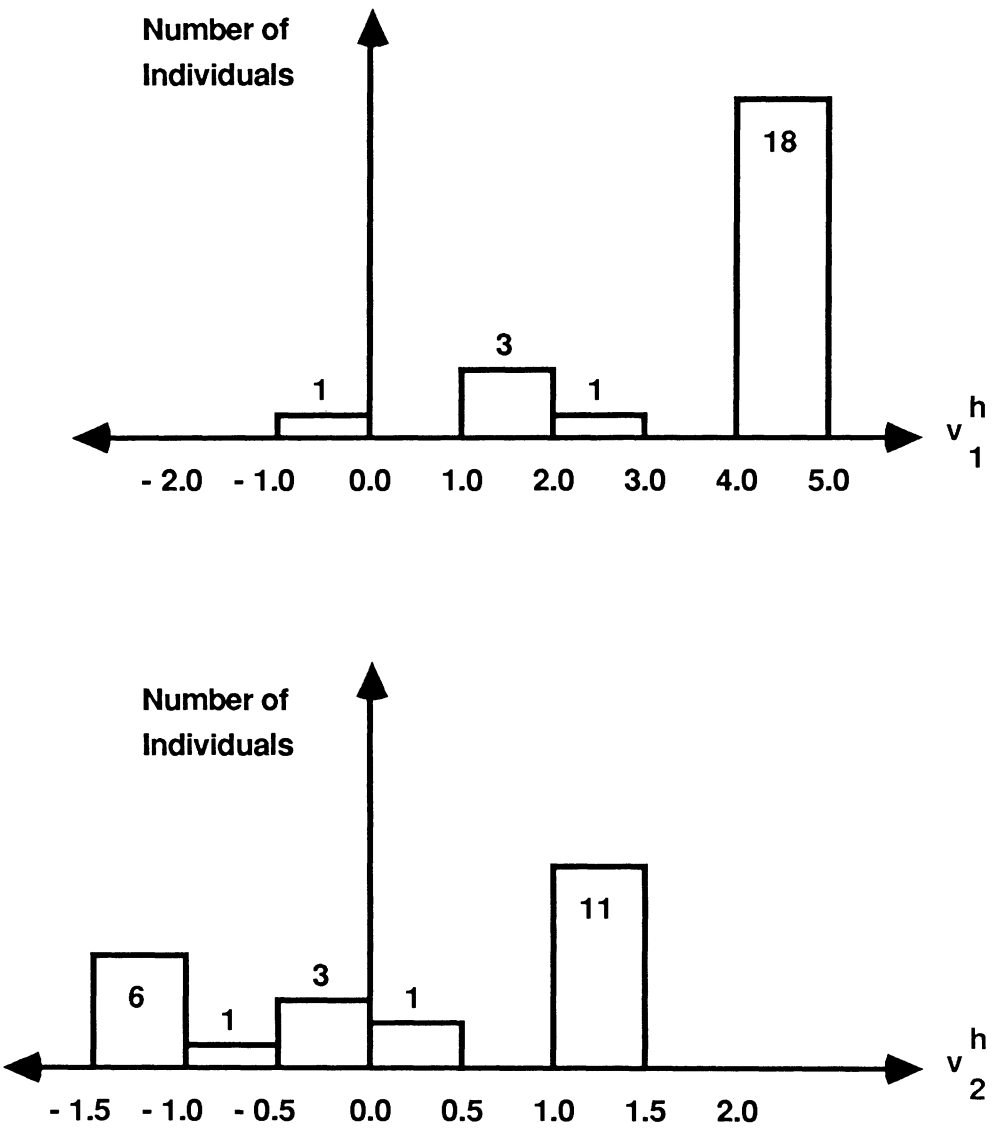


FIGURE 1. Distributions of CSB Model Parameters v_1^h and v_2^h .

In this illustration, the implications that stem from the balance model calibrated at the individual level are also apparent in the cross-sectional specification. Individual motivations for varied choice behavior appear to be attribute-specific. On the whole, individuals limit their choice to a certain set of brands, and exhibit a tendency to choose a brand that has been recently chosen. Most choose consistently either from among nondiet beverages or diet beverages (loyal behavior), but many (43%) are liable to choose a noncola to follow the recent consumption of a cola (desire for variety).

5. Conclusion

The results from the balance model suggest the importance of being able to identify attribute-level motivations for varied choice behavior. Marketing strategies should be targeted to guard jealously the established loyalties of consumers along certain dimen-

sions, and to exploit consumers' desires to find diversity along others. The existence of a variety-seeking segment in a product category indicates the possibility of a fruitful line-extension strategy; the information from the balance model can suggest the appropriate direction for such a strategy. In the future, as brands proliferate, it may become increasingly important to identify new products which serve to complement (rather than cannibalize) existing offerings. Pre-test market studies for new products might find it useful to employ the balance model to discover any tendencies for balanced consumption behavior within the household.

The results of the balance model can also have implications for marketing tactics. Cross-promotional offers (e.g., in which a special coupon for one brand appears on or in the package of another) may be influenced according to the variety-seeking propensities of consumer segments. Thus, it may be to Coke's advantage to offer coupons for Sprite (rather than Tab) on the label of its 2-liter bottle, since the principal competitive effect would fall on 7Up (or other noncola).

The balance model can be calibrated at the individual level (if individual choice histories are sufficiently long) or at the cross-sectional level. While there may be some inaccuracy introduced by the cross-sectional specification (see our comments in §4.2 above; see also the comments by Bass et al. 1984, p. 285), our validation suggests that the balance model still performs well in fitting and predicting choice. Nonetheless, these conjectures must remain tentative until the balance model can be tested on other more general sources of individual consumption data.¹

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